

Regression

1: Regression determines the relationship between independent variables and a continuous target variable. It identifies trends and patterns in data to make future predictions or estimate unknown values based on learned relationships.

- the difference between Classification and Regression. Both are supervised learning techniques, but they solve different types of problems depending on the nature of the target variable.

Regression	Classification
Predicts a continuous numeric value	Predicts a categorical label
Fitting the best-fit line or curve	Drawing a decision boundary between classes
Forecasting, price prediction, risk estimation	Image recognition, sentiment analysis, fraud detection

- Real-life example

Regression Example: Predicting the market value of a used car based on features like manufacturing year and engine capacity

Classification Example: predicts categories or labels like disease/no disease, Yes/NO.

2: **Linear Regression:** Fits a straight line to model the relationship between input features and a continuous output.

- Multiple linear Regression:** Extends linear regression by using multiple input variables for more accurate predictions.
- Polynomial Regression:** Models non-linear relationships by introducing polynomial terms of the input features.

Linear Regression	Multiple Linear Regression	Polynomial Regression
Advantage: Linear Regression is simple to implement and easier to interpret the output coefficients.	Advantage: Captures the combined influence of many factors simultaneously	Advantage: Effectively captures non-linear relationships without switching to non-linear algorithms
Disadvantage: On the other hand in linear regression technique outliers can have huge effects on the regression and boundaries are linear in this technique.	Disadvantage: Performance drops in the presence of multicollinearity (features highly correlated with each other)	Disadvantage: Higher-degree polynomials may lead to overfitting and unstable predictions
Use Case: Healthcare and Medical Research: Predict disease progression, patient outcomes, and treatment effectiveness using medical data.	Use Case: E-commerce: Analyze how price, promotions and seasons affect sales.	Use Case: Population growth: Predicting population growth over time

- **Mean absolute error:** represents the average of the absolute difference between the actual and predicted values in the dataset. It measures the average of the residuals in the dataset.
- **Mean Squared Error:** represents the average of the squared difference between the original and predicted values in the data set. It measures the variance of the residuals.
- **Root Mean Squared Error:** is the square root of Mean Squared error. It measures the standard deviation of residuals.
- **R-squared represents:** the proportion of the variance in the dependent variable which is explained by the linear regression model.

Metric	Units	Sensitivity to large Error/Outliers	Meaning
Mean absolute error	Same as target	Low	Average Absolute Prediction error
Mean Squared Error	Square unit	Very High	Average squared Prediction error
Root Mean Squared Error	Same as target	high	Measures the typical size of prediction error
R-squared	unitless	Moderate	shows how much of the variation in the target is explained by model(0-1; higher is better)

4: underfitting: happens when the model fails to learn important patterns. It performs poorly on both training and testing data.

Overfitting: happens when the model learns too much from the training data, including noise and outliers. It performs very well on training data but poorly on test data.
causes of overfitting in polynomial is primary caused by choosing an excessively high polynomial degree(n). when the degree is too high.

To prevent overfitting, the main objective is to make the model generalize well to new data instead of memorizing the training set. This can be achieved by controlling model complexity and improving data usage.

- **Cross Validation:** Instead of evaluating the model on just one split of data
- **Regularization (L1 / L2):** adds a penalty to the model's loss function. This discourages overly large coefficients and prevents the model from becoming too complex. L1 can remove less important features, while L2 reduces their influence.

5:Real-world case study

- Sales prediction for a small retail business
- The Goal of the project was to predict daily sales for a small retail business based on factors such as product price , number of customers and day type.
- dataset included: Product price customer per day, day type, product category, item sold
- Regression type: Multiple Linear Regression
- key results/insights: the model showed that the number of customers and the type of day had a strong effect on sales. weekend sales were generally higher than weekday sales the results helped the business estimate future sales and improve inventory planning